Student Management HTTP Web API

DATA2410 - Assignment #3

OsloMet - Faculty of Technology, Art and Design (TKD)

Date: 04.05.2026

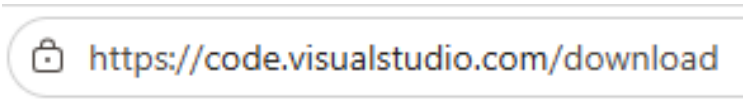
Contents

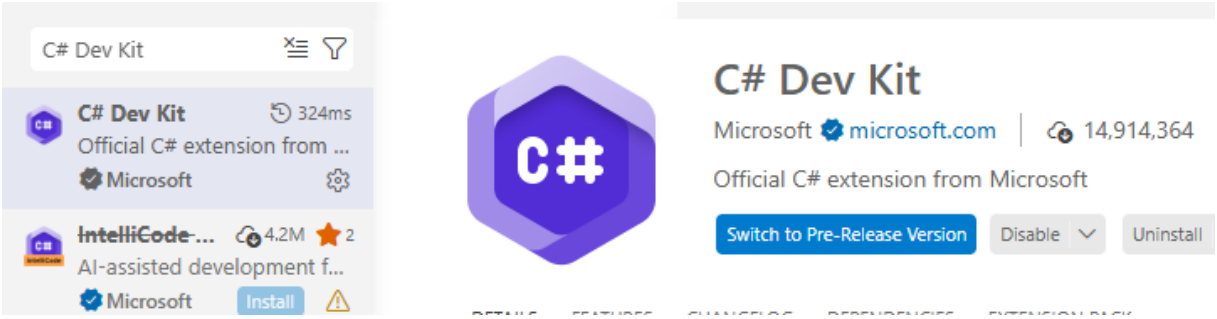
Project Setup: C# ASP.NET Core + SQL Server 1
1 INTRO 1
1.1 Install Visual Studio Code 1
1.2 Install Git 2
1.3 Clone the Repository 2
1.4 Install .NET 10 SDK 3
1.5 Install SQL Server and SSMS 4
1.5.1 SQL (Structured Query Language) SERVER INSTALL 4
1.5.2 SSMS (SQL Server Management Studio) INSTALL 5
1.6 Configure the Connection String 6
1.6.1 Option 1 (Windows Authentication) CHOSEN METHOD 6
1.6.2 Option 2 (SQL Server Authentication) 6
1.7 Run the Project 7

Project Setup: C# ASP.NET Core + SQL Server

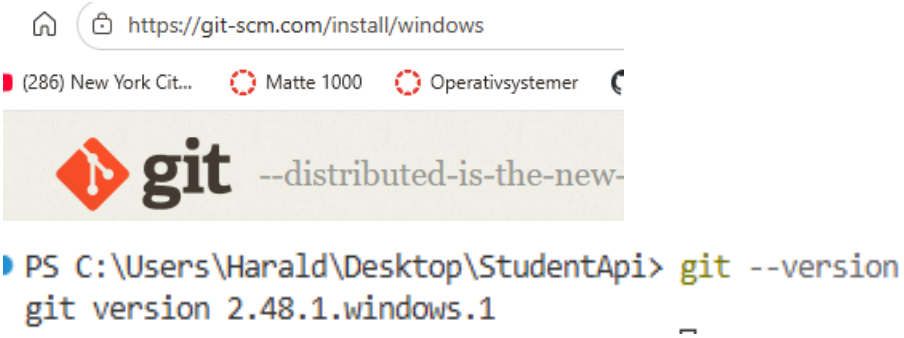
1 INTRO

1.1 Install Visual Studio Code

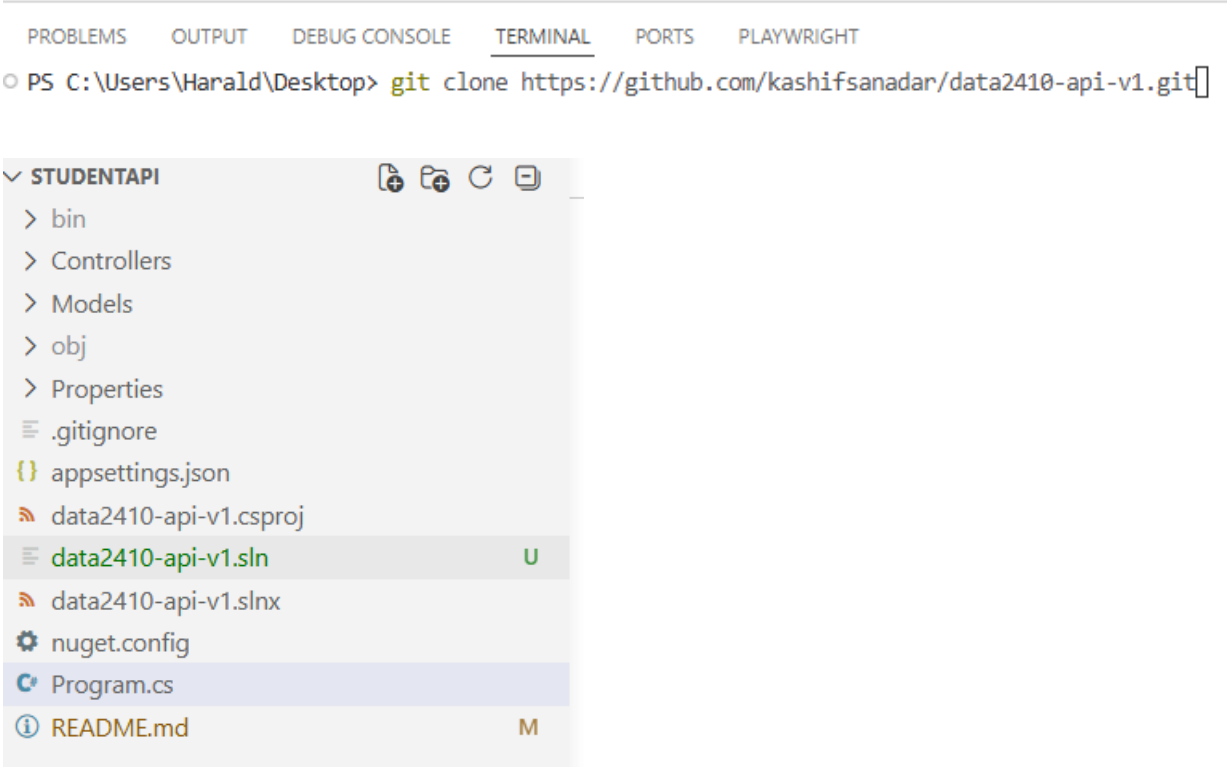




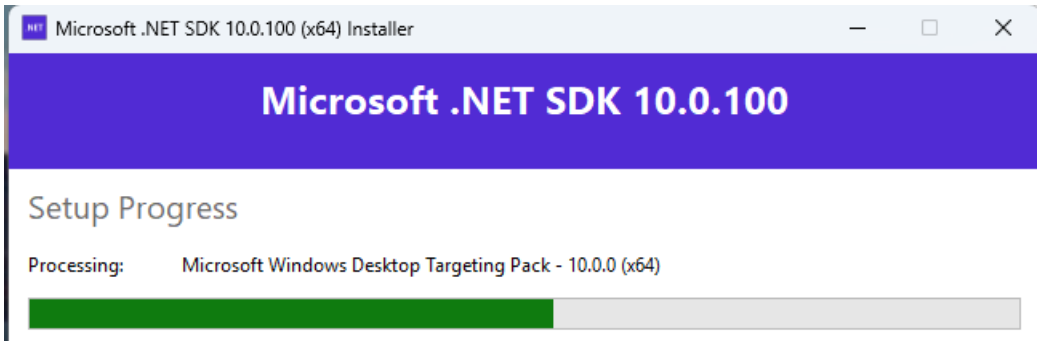
1.2 Install Git



1.3 Clone the Repository



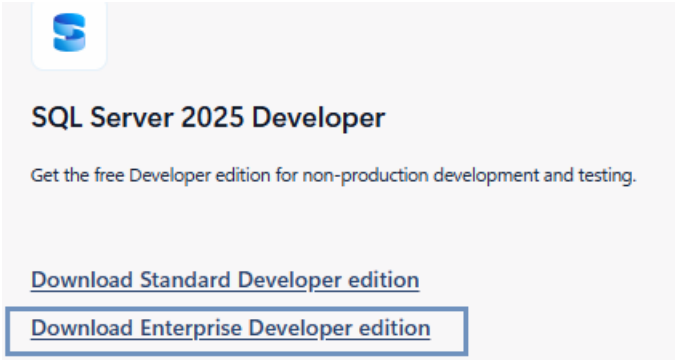
1.4 Install .NET 10 SDK




```
PS C:\Users\Harald\Desktop\StudentApi> dotnet --version
10.0.100
PS C:\Users\Harald\Desktop\StudentApi>
```

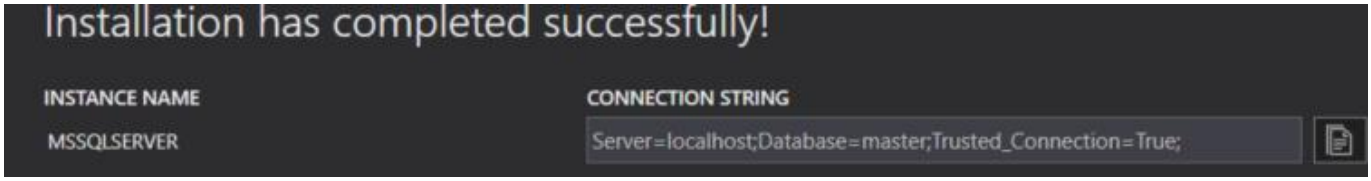
1.5 Install SQL Server and SSMS

1.5.1 SQL (Structured Query Language) SERVER INSTALL





4. During setup, note down the **server name** (usually `localhost`) and configure **SQL Server Authentication** with a username and password



1.5.2 SSMS (SQL Server Management Studio) INSTALL

https://learn.microsoft.com/en-us/ssms/install/install

Download SQL Server Management Studio 22 installer

SQL Server Management Studio 22

22.5.0

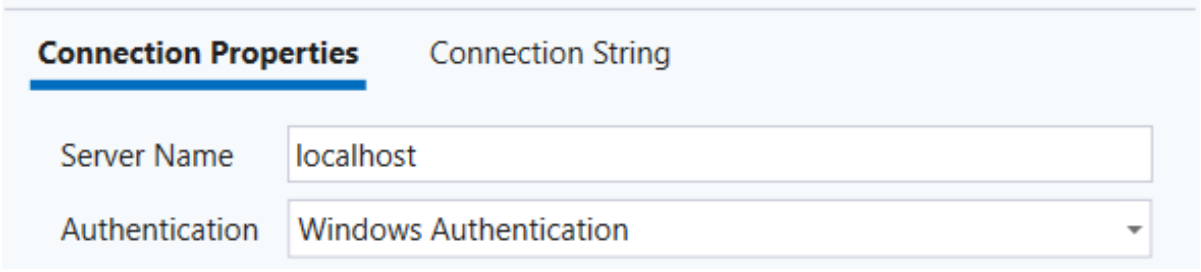
Release notes

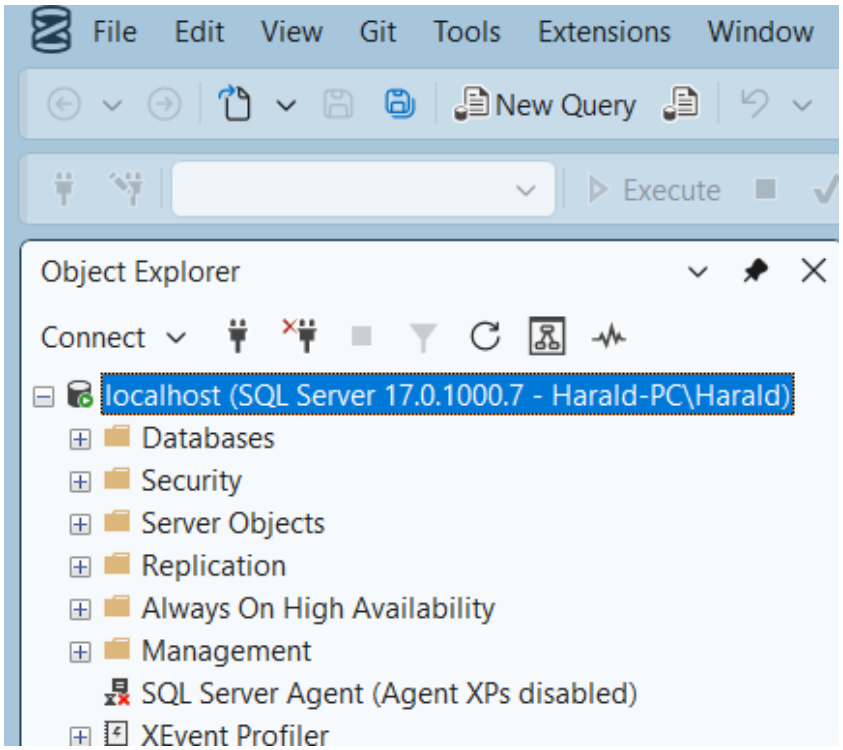
Modify

Launch

More

After installation, open SSMS and connect to your local SQL Server using the server name and credentials you configured above





SSMS Interface

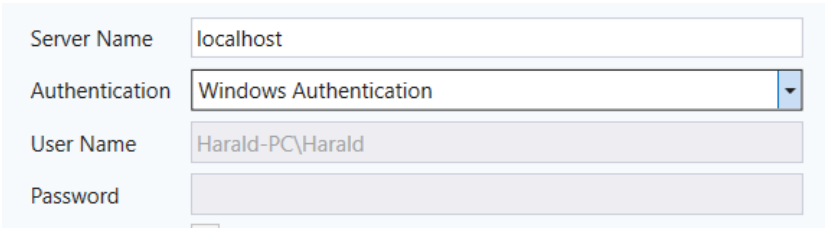
1.6 Configure the Connection String



File: appsettings.json

1.6.1 Option 1 (Windows Authentication) CHOSEN METHOD

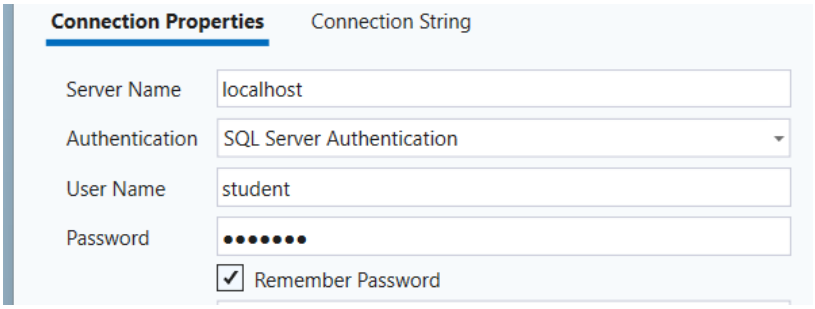
- Using Windows Authentication (no username/password).
- Removed User ID and Password from DefaultConnection.
- "DefaultConnection": "Server=localhost;Database=StudentsDb;User ID=sa;Password=YOUR_PASSWORD;TrustServerCertificate=Tru
- "DefaultConnection": "Server=localhost;Database=StudentsDb;User ID=sa;Password=YOUR_PASSWORD;TrustServerCertificate=True;"

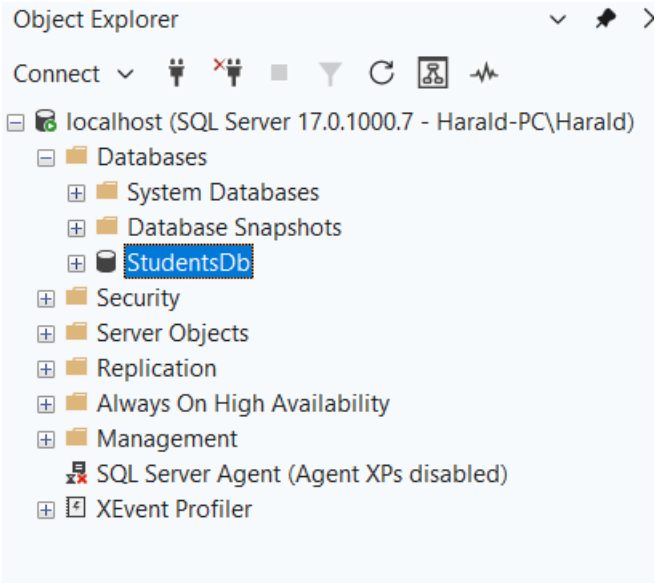


1.6.2 Option 2 (SQL Server Authentication)

- SQL Server Authentication, then add User ID and Password. In appsettings.json

"DefaultConnection": "Server=localhost;Database=StudentsDb;User ID=student;Password=student;TrustServerCertificate=True;"





Created Studentdb under Object Exploer > Databases

1.7 Run the Project

```
dotnet restore
dotnet run
```

```
cd "C:\Users\Harald\Desktop\S374171_A3_StudentApi"
dotnet restore .\data2410-api-v1.csproj
dotnet run --project .\data2410-api-v1.csproj
```

```
PS C:\Users\Harald\Desktop\S374171_A3_StudentApi> cd "C:\Users\Harald\Desktop\S374171_A3_StudentApi"
>> dotnet restore .\data2410-api-v1.csproj
>> dotnet run --project .\data2410-api-v1.csproj
Restore complete (0,9s)

Build succeeded in 1,6s
Using launch settings from .\Properties\launchSettings.json...
Building...
C:\Users\Harald\Desktop\S374171_A3_StudentApi\Models\Student.cs(9,19): warning CS8618: Non-nullable property 'Course' must contain a
 non-null value when exiting constructor. Consider adding the 'required' modifier or declaring the property as nullable.
C:\Users\Harald\Desktop\S374171_A3_StudentApi\Controllers\StudentsController.cs(79,22): warning CS8605: Unboxing a possibly null val
ue.
warn: Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServer[8]
      The ASP.NET Core developer certificate is not trusted. For information about trusting the ASP.NET Core developer certificate,
see https://aka.ms/aspnet/https-trust-dev-cert
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:7010
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\Harald\Desktop\S374171_A3_StudentApi
```